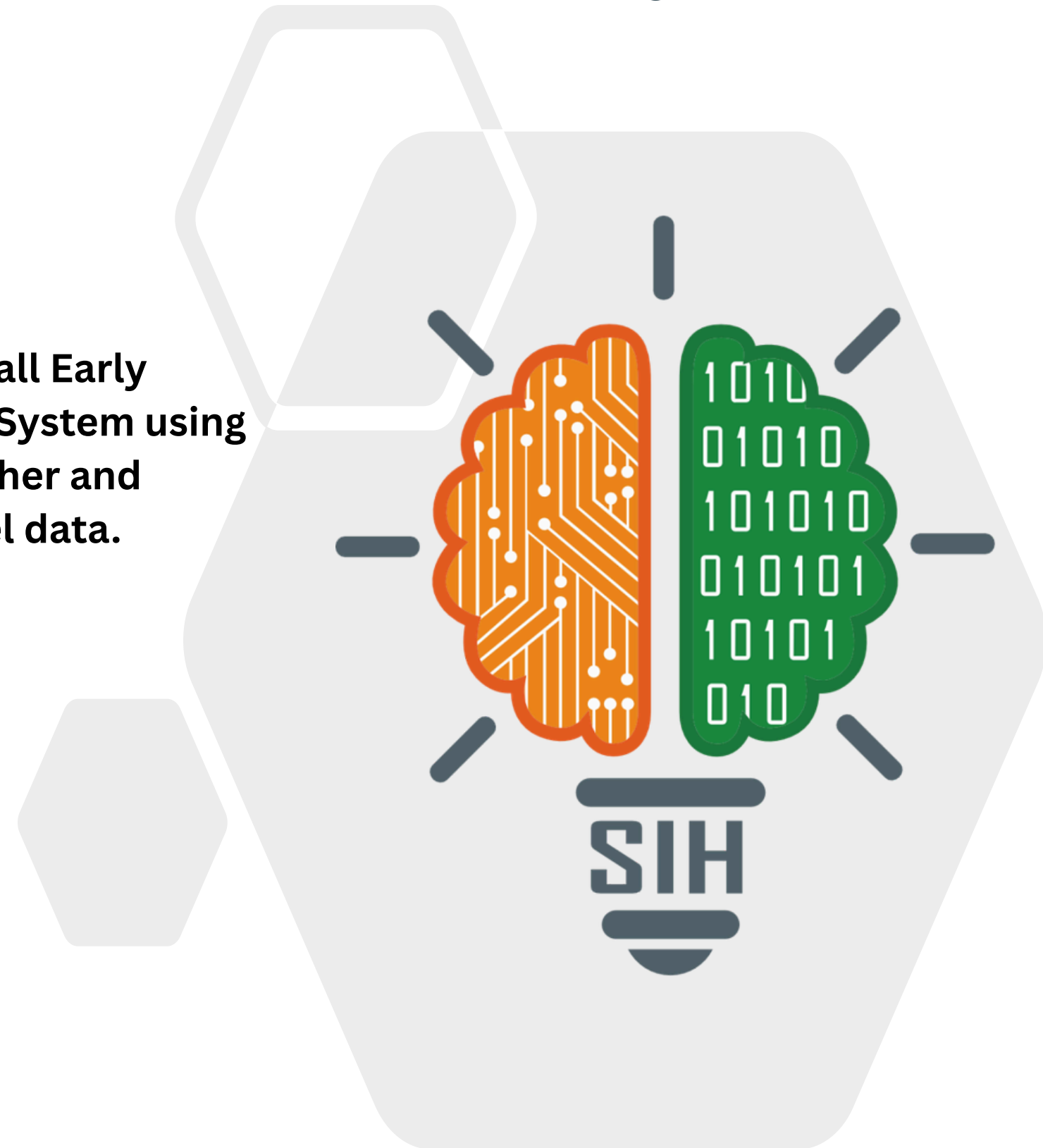


TITLE PAGE

- **Problem Statement ID – SIH26071**
- **Problem Statement Title-** AI/ML-Based Integrated heavy rainfall Early Warning and Inundation Prediction System using Satellite, Radar, observational Weather and numerical weather prediction model data.
- **Theme- Disaster Management**
- **PS Category- Software**
- **Team ID- 133517**
- **Team Name - SelfNProve**



HydroWatch fuses weather observations, radar, numerical forecasts and satellite imagery with two ML models to predict heavy rainfall, map water/inundation and produce explainable location-specific risk assessments.

THE GAPS

Fragmented signals

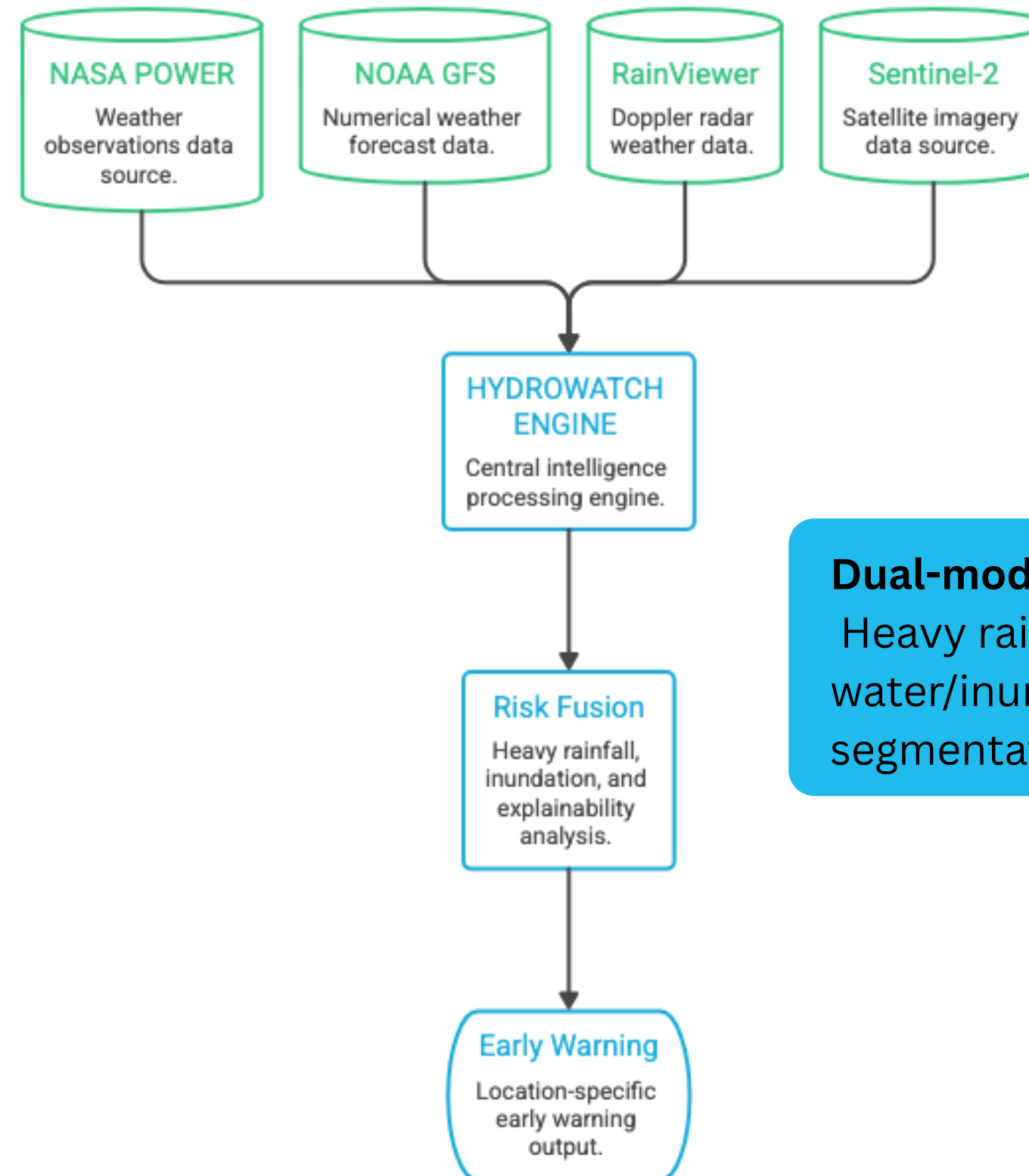
Weather, radar, satellite and forecasts are usually analysed separately.

Limited spatial context

Rainfall risk alone does not show where water may accumulate.

Decision uncertainty

Different data sources vary in timing, scale and reliability.



WHAT MAKES IT DIFFERENT

Multi-source fusion

Observations + radar + NWP + satellite

Dual-model intelligence

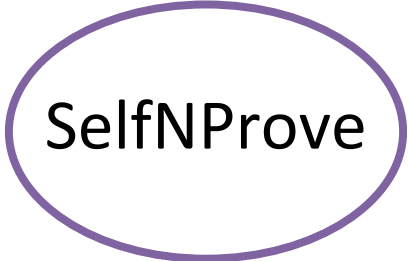
Heavy rainfall + water/inundation segmentation

Explainable decisions

Risk contribution from each evidence stream

Geospatial output

Actual location-specific polygons on a 3D map

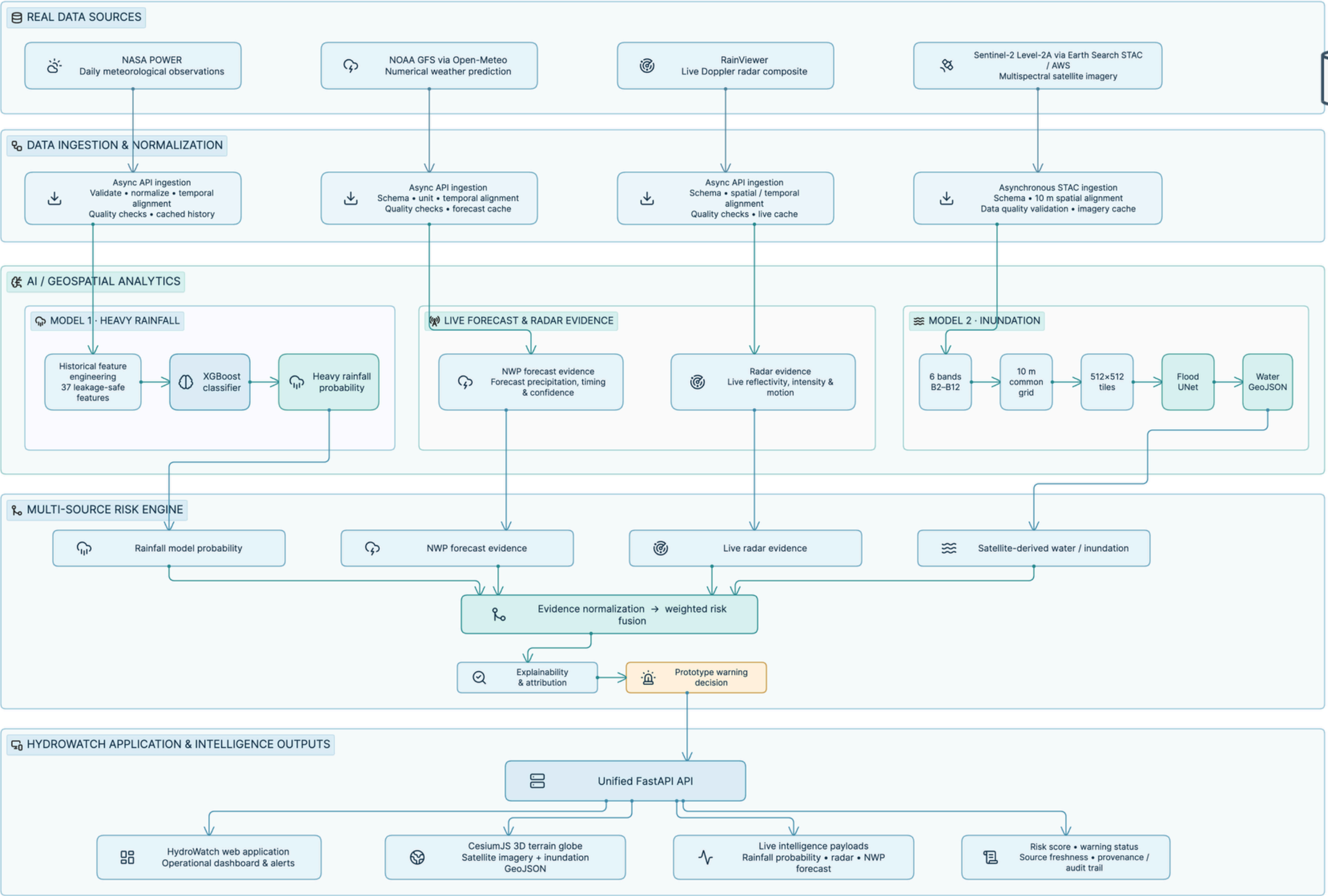


TECHNICAL APPROACH

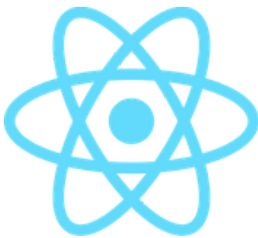
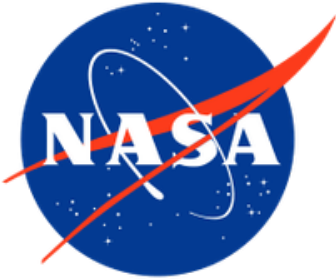


SMART INDIA
HACKATHON
2026

Enterprise layered architecture - multi-source weather, radar, satellite, AI and decision intelligence



TECH STACK

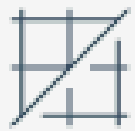
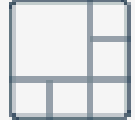


REAL DATA
Public machine-readable weather, radar, forecast and satellite sources.

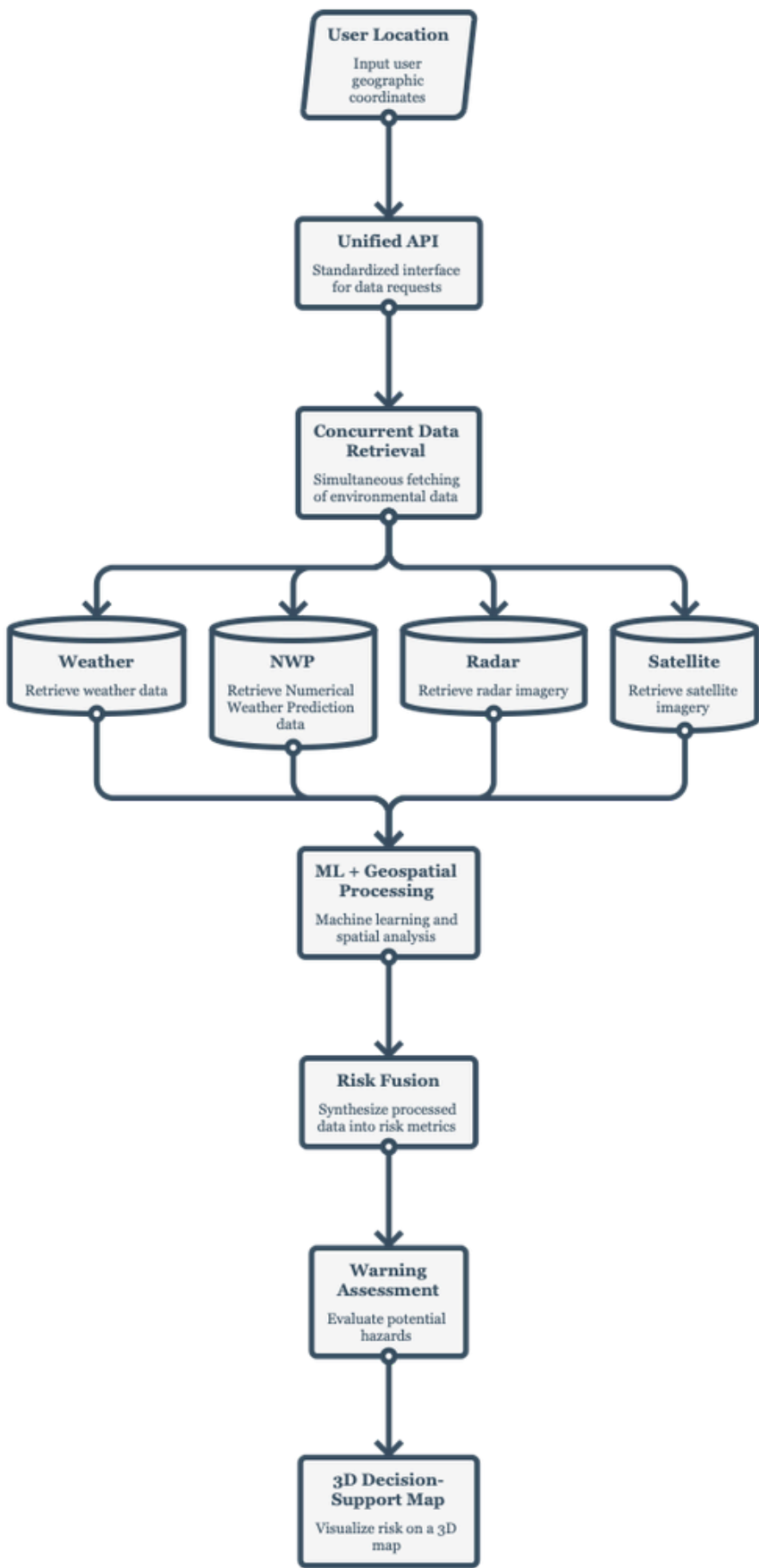
MODULAR
Each source, model and service can be replaced independently.

COMPUTATIONALLY PRACTICAL
Tiled satellite inference + cached external feeds + singleton ML models.

DEPLOYABLE
FastAPI backend + Next.js frontend + container-ready architecture.

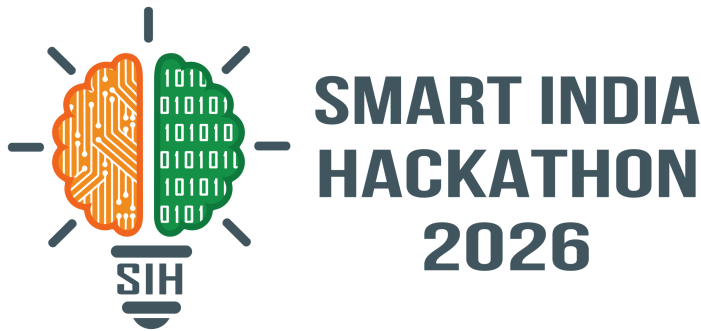
Challenge	Engineering Response
 Different spatial resolutions	Common geospatial grid
 Different timestamps	Temporal alignment
 Large satellite rasters	512×512 tiled inference
 Missing feeds	Explicit partial-source policy
 Historical satellite scenes	Freshness / temporal labeling
 Provider failures	Structured upstream errors

Geospatial Risk Assessment Workflow





IMPACT AND BENEFITS

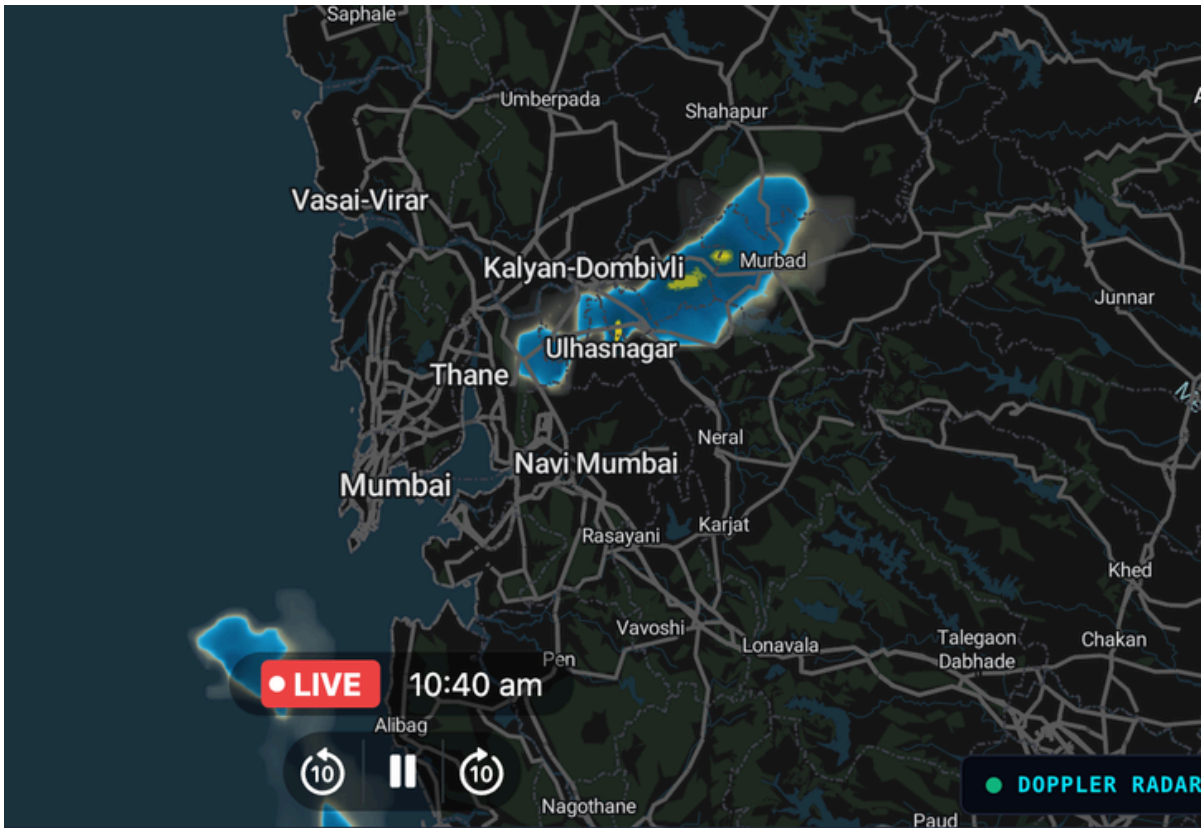
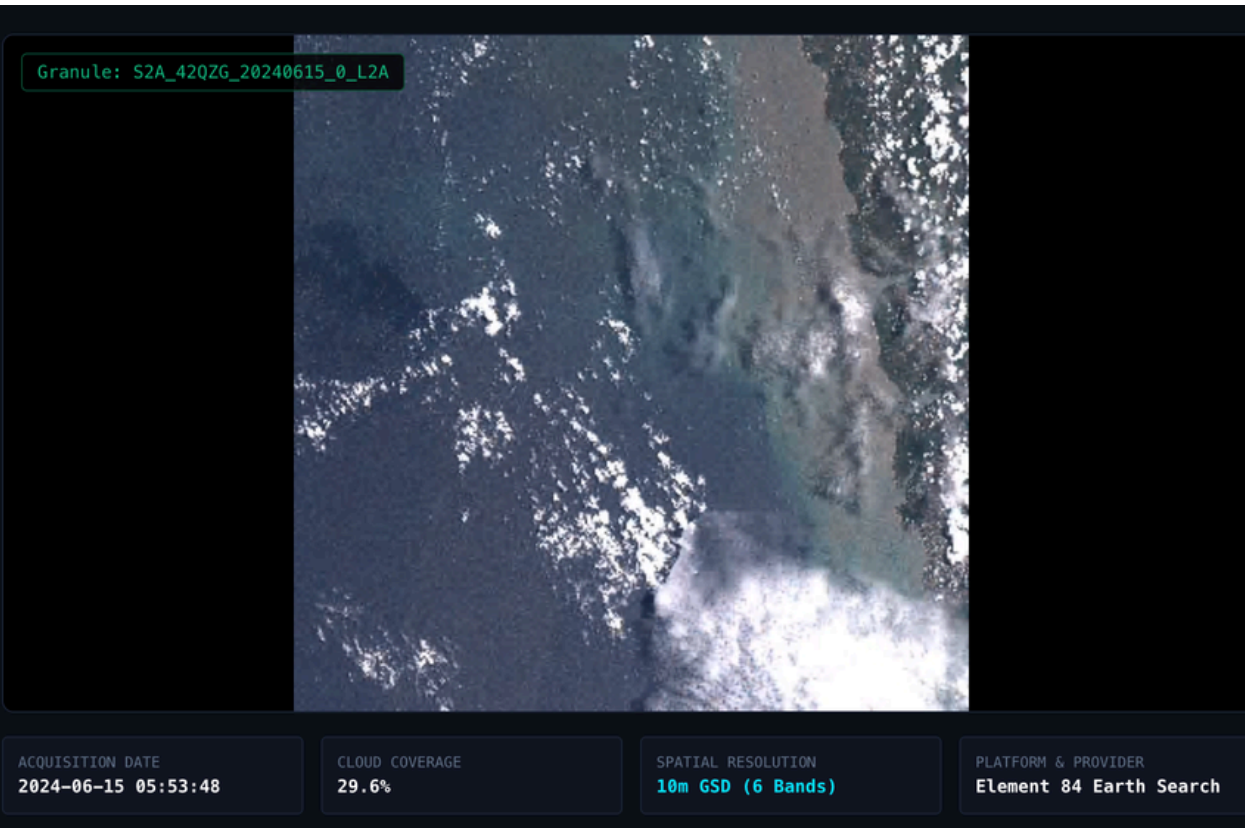
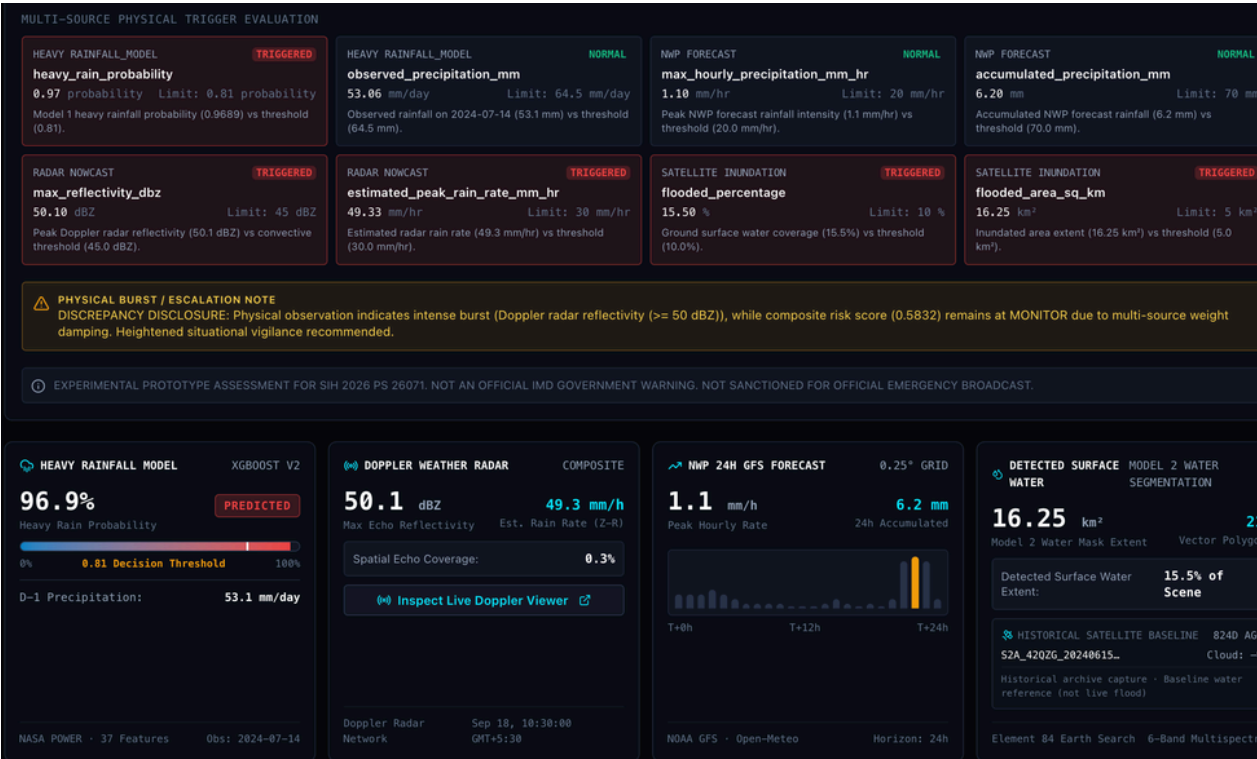
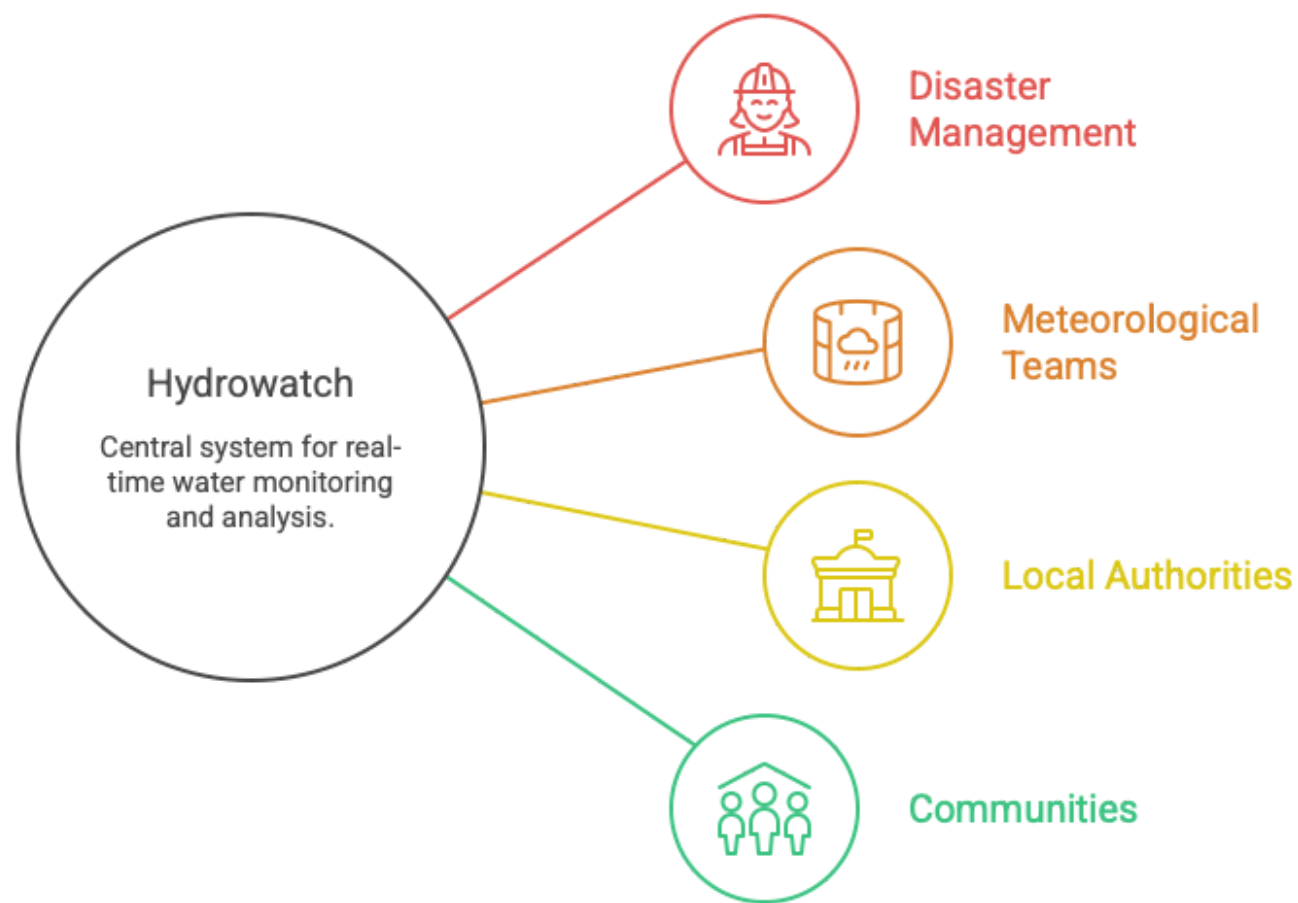


EARLIER AWARENESS
Location-specific environmental signals

SPATIAL DECISION SUPPORT
Actual inundation extent on map

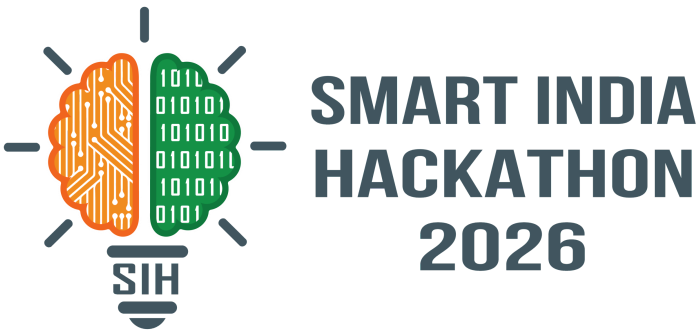
EXPLAINABLE INTELLIGENCE
Traceable evidence behind the assessment

Unveiling Hydrowatch's Multifaceted Impact





RESEARCH AND REFERENCES



DATA & EARTH OBSERVATION

NASA POWER

Daily meteorological observations used for rainfall-model features.

[NASA POWER](#)

Sentinel-2 Level-2A

Multispectral surface-reflectance imagery for water/inundation segmentation.

[SENTINEL 2 DATA](#)

Earth Search STAC

Programmatic Sentinel-2 scene discovery and metadata.

[EARTH SEARCH](#)

AWS Open Data

Cloud-hosted Sentinel-2 geospatial assets.

[AWS OPEN DATA](#)

WEATHER & RADAR

NOAA GFS via Open-Meteo

Hourly numerical weather prediction.

[Open-Meteo GFS](#)

RainViewer

Live radar composite observations and radar-map tiles.

[RainViewer API](#)

[RainViewer Weather Maps API](#)

Weather observations → Forecast → Radar

VERIFIED MODEL PERFORMANCE

HEAVY RAINFALL PREDICTION

TEST ACCURACY : 92.27%

ROC-AUC · 92.28%

WATER / INUNDATION SEGMENTATION

PIXEL ACCURACY :

97.79%

Dice / F1 : 91.21%

DOC : [Comprehensive Doc](#)

ML/MODELS

XGBoost

Heavy-rainfall classification from 37 engineered features.

[XG-BOOST](#)

PyTorch / FloodUNet

Six-band satellite water segmentation.

[PYTORCH](#)

SHAP

Model feature attribution for rainfall prediction.

[SHAP DOC](#)

Sen1Floods11 v1.1

Flood-segmentation training/reference dataset.

[Sen1Floods11 v1.1](#)

Metrics are reported on the held-out test sets used during model development.